

CLAUDE

INHERITED FROM Helix Constitution

This module is a submodule of an ATMOSphere-family project that includes the Helix Constitution submodule at the parent's constitution/ path. All rules in constitution/CLAUDE.md and the constitution/Constitution.md it references (universal anti-bluff covenant §11.4, no-guessing mandate §11.4.6, credentials-handling mandate §11.4.10, host-session safety §12, data safety §9, mutation- paired gates §1.1) apply unconditionally to every change landed here. The module-specific rules below extend them — they never weaken any universal clause.

When this file disagrees with the constitution submodule, the constitution wins. Locate the constitution submodule from any arbitrary nested depth using its find_constitution.sh helper.

Canonical reference: <https://github.com/HelixDevelopment/Helix-Constitution>

CLAUDE.md - ATMOSphere Gramophone Player

Fork of Gramophone music player, customized for ATMOSphere firmware on Orange Pi 5 Max (RK3588). Material Design 3 music player with Media3/ExoPlayer backend and hi-fi audio support.

Project Overview

- **Package:** atmosphere.gramophone.player
- **Language:** Kotlin
- **Build:** Gradle KTS (AGP 9.0.1, Kotlin 2.3.0)
- **Repo:** git@github.com:ATMOSphere1234321/ATMOSphere-Gramophone-Player.git
- **Parent repo path:** device/rockchip/atmosphere/gramophone-player

- **Upstream:** AkaneTan/Gramophone

Key Directories

Directory	Purpose
app/	Main Android app module
media3/	Media3/ExoPlayer integration
hificore/	Hi-fi audio processing core
baselineprofile/	Startup and runtime performance profiles

ATMOSphere Integration

- Listed in `AUDIO_ONLY_PACKAGES` -- excluded from video routing to secondary display
- Uses Media3/ExoPlayer for audio playback (MediaSession published)
- Audio output routed through ES8388 HAL (headphone/speaker) and BT A2DP
- CPU2 audio isolation applies to audioserver serving this app

MANDATORY HOST-SESSION SAFETY (Constitution §12)

Forensic incident, 2026-04-27 22:22:14 (MSK): the developer's `user@1000.service` was SIGKILLED under an OOM cascade triggered by `pip3 install --user openai-whisper` running on top of chronic podman-pod memory pressure. The cascade SIGKILLED `gnome-shell`, every `ssh` session, `claude-code`, `tmux`, `htop`, `npm`, `node`, `java`, `pip3` — full session loss. Evidence: `journalctl --since "2026-04-27 22:00" --until "2026-04-27 22:23"`.

This invariant applies to **every script, test, helper, and AI agent** in this submodule. Non-compliance is a release blocker.

Forbidden — directly OR indirectly

1. **Suspending the host:** `systemctl suspend`, `pm-suspend`, `loginctl suspend`, `DBus org.freedesktop.login1.Suspend`, `GNOME idle-suspend`, `lid-close handler`.
2. **Hibernating / hybrid-sleeping:** any `Hibernate` / `HybridSleep` / `SuspendThenHibernate` method.
3. **Logging out the user:** `loginctl terminate-session`, `kill -u <user>`, `systemctl --user --kill`, anything that signals `user@<uid>.service`.
4. **Unbounded-memory operations** inside `user@<uid>.service` cgroup. Any single command expected to exceed 4 GB RSS

MUST be wrapped in `bounded_run` (defined in `scripts/lib/host_session_safety.sh`, parent repo).

5. **Programmatic rkill toggles, lid-switch handlers, or power-button handlers** — these cascade into idle-actions.
6. **Disabling systemd-logind, GDM, or session managers** "to make things faster" — even temporary stops leave the system unable to recover the user session.

Required safeguards

Every script in this submodule that performs heavy work (build, transcription, model inference, large compression, multi-GB git op) MUST:

1. Source `scripts/lib/host_session_safety.sh` from the parent repo.
2. Call `host_check_safety` at the top and **abort if it fails**.
3. Wrap any subprocess expected to exceed ~4 GB RSS in `bounded_run "<name>" <max-mem> <max-time> -- <cmd...>` so the kernel OOM killer is contained to that scope and cannot escalate to `user.slice`.
4. Cap parallelism (`-j`) to fit available RAM (each AOSP job $\approx$ 5 GB peak RSS).

Container hygiene

Containers (Docker / Podman) we own or rely on MUST:

1. Declare an explicit memory limit (`mem_limit / --memory / MemoryMax`).
2. Set `OOMPolicy=stop` in their `systemd` unit to avoid retry loops.
3. Use exponential-backoff restart policies, never immediate retry.
4. Be clean-slate destroyed (`podman pod stop && rm, podman volume prune`) and rebuilt after any host crash or session loss so stale lock files don't keep producing failures.

When in doubt

Don't run heavy work blind. Check `journalctl -k --since "1 hour ago" | grep -c oom-kill`. If it's non-zero, **fix the offending workload first**. Do not stack new work on a host already in distress.

Cross-reference: parent `docs/guides/ATMOSPHERE_CONSTITUTION.md` §12 (full forensic, library API, operator directives) + parent `scripts/lib/host_session_safety.sh`.

MANDATORY ANTI-BLUFF VALIDATION (Constitution §8.1 + §11)

This submodule inherits the parent ATMOSphere project's anti-bluff covenant. A test that PASSES while the feature it claims to validate is unusable to an end user is the single most damaging failure mode in this codebase. It has shipped working-on-paper / broken-on-device builds before, and that MUST NOT happen again.

The canonical authority is docs/guides/ATMOSPHERE_CONSTITUTION.md §8.1 ("NO BLUFF — positive-evidence-only validation") and §11 ("Bleeding-edge ultra-perfection") in the parent repo. Every contribution to THIS submodule is bound by it. Summarised non-negotiables:

1. **Tests MUST validate user-visible behaviour, not just metadata.** A gate that greps for a string in a config XML, an XML attribute, a manifest entry, or a build-time symbol is METADATA — not evidence the feature works for the end user. Such a gate is allowed ONLY when paired with a runtime / on-device test that exercises the user-visible path and reads POSITIVE EVIDENCE that the behaviour actually occurred (kernel /proc/* runtime state, captured audio/video, dumphsys output produced *during* playback, real input-event delivery, real surface composition, etc).
2. **PASS / FAIL / SKIP must be mechanically distinguishable.** SKIP is for environment limitations (no HDMI sink, no USB mic, geo-restricted endpoint unreachable) and MUST always carry an explicit reason. PASS is reserved for cases where positive evidence was observed. A test that completes without observing evidence MUST NOT report PASS.
3. **Every gate MUST have a paired mutation test in scripts/testing/meta_test_false_positive_proof.sh (parent repo).** The mutation deliberately breaks the feature and the gate MUST then FAIL. A gate without a paired mutation is a BLUFF gate and is a Constitution violation regardless of how many checks it appears to make.
4. **Challenges (HelixQA) and tests are in the same boat.** A Challenge that reports "completed" by checking the test runner exited 0, without observing the system behaviour the Challenge is supposed to verify, is a bluff. Challenge runners MUST cross-reference real device telemetry (logcat, captured frames, network probes, kernel state) to confirm the user-visible promise was kept.
5. **The bar for shipping is not "tests pass" but "users can use the feature."** If the on-device experience does not match what the test claims, the test is the bug. Fix the test (positive-evidence harder), do not silence it.
6. **No false-success results are tolerable.** A green test suite combined with a broken feature is a worse outcome than an

honest red one — it silently destroys trust in the entire suite. Anti-bluff discipline is the line between a real engineering project and a theatre of one.

When in doubt: capture runtime evidence, attach it to the test result, and let a hostile reviewer (i.e. yourself, in six months) try to disprove that the feature really worked. If they can, the test is bluff and must be hardened.

Cross-references: parent CLAUDE.md "MANDATORY DEVELOPMENT PRINCIPLES", parent AGENTS.md "NO BLUFF" section, parent scripts/testing/meta_test_false_positive_proof.sh.

MANDATORY DEVELOPMENT PRINCIPLES

1. **Solutions MUST NOT be error-prone** -- every fix must be robust, not introduce new failure modes
2. **No blocking operations inside synchronized blocks** -- Thread.sleep(), network calls, or long computations inside synchronized WILL cause deadlocks
3. **Always consider concurrent callers** -- multiple media sessions can be active simultaneously
4. **Test the fix, not just the symptom** -- verify the fix works AND does not break anything else

MANDATORY API KEY & SECRETS CONSTRAINTS

1. **NEVER commit .env files** -- they contain API keys and credentials
2. **NEVER add API keys to source code** -- use environment variables or .env files only
3. **ALWAYS verify .gitignore protects .env** before committing

MANDATORY COMMIT & PUSH CONSTRAINTS

1. **ONLY use the official commit script from the PARENT repo:** bash scripts/commit_all.sh "message"
2. **NEVER use git add, git commit, or git push directly** in this submodule
3. The parent script handles staging, committing, and pushing to ALL remotes

MANDATORY SUBMODULE SYNC CONSTRAINTS

1. **ALWAYS fetch and pull latest from upstream** before pushing our committed changes
2. **Analyze all new features/APIs** from upstream and incorporate them properly
3. **Merge conflicts** must be resolved carefully -- never discard upstream changes blindly

MANDATORY TAGGING CONSTRAINTS

1. **Tags are NEVER created before flashing and validating** on BOTH ATMOSphere devices
2. **Tags MUST be applied to ALL owned submodules** when tagging the main repo
3. **Tag naming:** <major>.<minor>.<patch>-dev[-<sub-version>]

Project Context

- Part of ATMOSphere Android 15 firmware for Orange Pi 5 Max (RK3588)
 - Parent repo at /run/media/milosvasic/DATA4TB/Projects/Android_15/ handles build, flash, and test
 - Build via parent: `bash scripts/build.sh --skip-pull --skip-tests --skip-ota`
 - Tests via parent: `bash device/rockchip/rk3588/tests/pre_build_verification.sh`
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MANDATORY: ATMOSphere Constitution compliance (appended 2026-04-19 — ATMOSphere 1.1.3-dev-0.0.6)

Every change in this submodule **MUST** comply with the canonical Constitution at docs/guides/ATMOSPHERE_CONSTITUTION.md in the parent repo. In summary:

1. **Test coverage for every change** — pre-build gate (CM-MC*), post-build gate, on-device test, and a mutation entry in `scripts/testing/meta_test_false_positive_proof.sh` proving the gate catches regressions.
2. **Device validation before any tag** — both D1 and D2 flashed and green on every relevant suite.

3. **Commit + push via bash scripts/commit_all.sh "... from the parent repo root.** Submodule source changes are committed in the submodule itself first, pushed to every remote of that submodule, and then the parent's commit_all.sh captures the updated pointer.
4. **Tags cascade.** Every version tag on the main repo is mirrored on this submodule at its current HEAD, across every remote this submodule publishes to. Use scripts/testing/release_tag.sh <tag> from the parent repo.
5. **Changelog discipline.** docs/changelogs/<tag>.{md,html,json,txt} on the parent repo documents every release; exported via scripts/testing/export_changelog.sh.
6. **No false-success results.** Tests that are always-PASS are immediately rewritten. Meta-test mutations catch bluff gates.
7. **Flock.** commit_all.sh and push_all.sh are serialised via .git/.commit_all.lock / .git/.push_all.lock. Never bypass.

Non-compliance is a blocker regardless of context.

MANDATORY ABSOLUTE DATA SAFETY — ZERO RISK (Constitution §9)

EVERY destructive repository operation (history rewrite, force-push, branch deletion, bulk file removal, submodule de-init, object pruning) **MUST** follow Constitution §9 without exception:

1. **Hardlinked backup first** — near-instant (cp -al .git <backup>/repo.git.mirror), zero extra disk. No excuse to skip. Parent-repo helper: scripts/testing/safe_history_rewrite.sh --pre-op.
2. **Record pre-op metadata** — refs, tags, submodule pointers, HEAD commit, HEAD tree hash, HEAD tree content sha256.
3. **Run the operation** — never with hook-bypassing flags unless the user has explicitly authorized them for that exact operation.
4. **Post-op gate** — HEAD tree byte-identical (unless explicitly expected to change), all tags preserved, all submodule pointers intact, all domain-specific integrity checks pass. ANY failure → restore immediately from the hardlinked backup.
5. **Force-push is NEVER automatic** — push_all.sh (in parent repo) must not force-push as a failure-recovery path. Every force-push requires explicit per-session human authorization AND a passing §9 post-op gate.
6. **Audit trail** — force-push events recorded in parent repo docs/changelogs/<tag>.md.

Data-safety violations are catastrophic (irreversible once the remote GCs dangling objects) and block the release cycle until fully remediated.

MANDATORY ANTI-BLUFF COVENANT — END-USER QUALITY GUARANTEE (User mandate, 2026-04-28)

Forensic anchor — direct user mandate (verbatim):

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

This is the historical origin of the project's anti-bluff covenant. Every test, every Challenge, every gate, every mutation pair exists to make the failure mode (PASS on broken-for-end-user feature) mechanically impossible.

Operative rule: the bar for shipping is **not** "tests pass" but **"users can use the feature."** Every PASS in this codebase MUST carry positive evidence captured during execution that the feature works for the end user. Metadata-only PASS, configuration-only PASS, "absence-of-error" PASS, and grep-based PASS without runtime evidence are all critical defects regardless of how green the summary line looks.

Tests AND Challenges (HelixQA) are bound equally — a Challenge that scores PASS on a non-functional feature is the same class of defect as a unit test that does. Both must produce positive end-user evidence; both are subject to the §8.1 five-constraint rule and §11 captured-evidence requirement.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §8.1 (positive-evidence-only validation) + §11 (bleeding-edge ultra-perfection quality bar) + §11.3 (the "no bluff" CLAUDE.md / AGENTS.md mandate) + **§11.4 (this end-user-quality-guarantee forensic anchor — propagation requirement enforced by pre-build gate CM-COVENANT-PROPAGATION).**

§11.4.1 extension (Phase 33, 2026-05-05) — FAIL-bluffs equally forbidden. A test that crashes for a script-internal reason (undefined variable under set -u, regex error, malformed assertion, missing argument) and produces a FAIL exit code is just as misleading as a PASS-bluff. Both let real defects ship undetected. Per parent [Constitution §11.4.1](#), every test MUST fail ONLY for genuine product defects — script-bug failures must be fixed at the source layer (helper library, shared lib, test source), not patched in individual call sites.

Non-compliance is a release blocker regardless of context.

§11.4.2 extension (Phase 34, 2026-05-06) — Recorded-evidence requirement. A test that emits PASS without captured visual or audio evidence of the user-visible feature actually working on the screen the user would see is a §11.4 PASS-bluff. Bug #13 (VK Video on PRIMARY display while a passing test claimed playback PASS) demonstrated the gap exactly. Closing it requires the recording + analyzer infrastructure (Bug #14 — `dual_display_record.sh / action_timeline.sh / Go recording-analyzer / helixqa-bridge`). Per Constitution §11.4.2 every PASS for a user-visible feature MUST be cross-checked by the analyzer against the dual-display recording + action timeline. A PASS that lacks at least one matched timeline event in the analyzer findings is treated as a §11.4 PASS-bluff.

Non-compliance is a release blocker regardless of context.

§11.4.3 extension (Phase 34, 2026-05-06) — Per-device-topology test dispatch. Tests that depend on hardware topology (secondary HDMI present/absent, microphone present/absent, etc.) MUST detect topology at test entry and dispatch the topology-appropriate variant. A test running the wrong variant for the actual topology and PASSing is a §11.4 PASS-bluff. Bug #18 (Lampa+TorrServe E2E) demonstrated the pattern: D1 (secondary HDMI) and D2 (primary only) get separate test variants behind a `dumpsys display-based` dispatcher. Per Constitution §11.4.3 every topology-touching test MUST have such a dispatcher OR explicit topology gates with SKIP-with-reason fallback.

Non-compliance is a release blocker regardless of context.

§11.4.4 extension (User mandate, 2026-05-06) — Test-interrupt-on-discovery + retest-from-clean-baseline. A test cycle that continues running past a freshly discovered defect is itself a §11.4 PASS-bluff: it produces "all green" summaries while the codebase under test is known-broken at the moment those greens were recorded. Phase 34.S' D1 demonstrated the violation when Bug #26 (hard-floor probe lifecycle) and Bug #27 (analyzer FAIL-bluff on non-video tests) were discovered mid-cycle and the cycle was allowed to continue, accumulating 13+ false-positive ANALYZER FAIL banners. Per Constitution §11.4.4 the moment any defect is re-discovered, re-produced, or newly identified during a test cycle, the cycle MUST stop on both devices. **Then:** (1) fix at root cause per §11.4.1, (2) land validation/verification tests for the fix — pre-build gate AND on-device test AND paired meta-test mutation, (3) full rebuild via `scripts/build.sh` (regardless of whether the fix touched host script / Go binary / firmware — host-only fixes still get a full rebuild for retest baseline integrity), (4) re-flash D1 + D2, (5) repeat full `test_all_fixes.sh` from the beginning sequentially per §12.6, (6) end the cycle with `meta_test_false_positive_proof.sh` proving no gate is itself a bluff

gate. Tests AND HelixQA Challenges are bound equally — Challenges that score PASS on a non-functional feature are the same class of defect as PASS-bluff unit tests; both must produce positive end-user evidence per §11.4.2 + §11.4.3.

Non-compliance is a release blocker regardless of context.

§11.4.4 expansion (User mandate, 2026-05-06) — Systematic debugging + four-layer test coverage + documentation + no-bluff certification. Augments the §11.4.4 base covenant with four non-negotiable additional requirements per the User mandate of 2026-05-06: (a) **Systematic debugging via superpowers skills.** Before applying any fix, run in-depth systematic debugging using the available superpowers:* skills (debugging, root-cause analysis, architectural-impact). Symptom patches are forbidden. The debugging output MUST identify root cause at source layer, blast radius across related tests/features/subsystems, and the regression-protection seam. (b) **Four-layer test coverage per fix.** Every fix lands with positive evidence in **every applicable layer**: pre-build gate (catches at source), post-build gate (catches in assembled image — proves bytes landed, cf. Fix #122 APK_LIB_MAP misroute), post-flash on-device test (fully automated, anti-bluff per §8.1, captured- evidence per §11.4.2, topology-dispatched per §11.4.3, orchestrator- wired in test_all_fixes.sh), HelixQA test bank entry (banks/atmosphere.yaml + per-feature additions), HelixQA full QA session coverage (Challenge-driven dispatch — bank entry without Challenge coverage is a §11.4 PASS-bluff), and meta-test paired mutation. Skipping a layer because "this fix only touches X" is forbidden. (c) **Documentation update for every fix.** Required: docs/Issues.md → docs/Fixed.md migration on closure, parent CLAUDE.md Applied Fixes Reference row, affected user-facing guides (docs/guides/*.md), affected diagrams/flowcharts/architecture docs, per-version docs/changelogs/<tag>.md entry. Documentation drift after a fix is itself a §11.4 violation. (d) **No-bluff certification per cycle.** Before tagging: meta_test_false_positive _proof.sh returns all gates green AND every gate's paired mutation FAILs (no bluff gates); docs/Issues.md open-set is empty or every entry explicitly classified out-of-scope-for-this-tag with operator sign-off (no known issues hidden); full suite returns zero new FAILs on either device (no working feature regressed); every gate has a paired mutation; every test produces positive evidence; every assertion catches its own negation (no error-prone or bluff-proof leftover).

Non-compliance is a release blocker regardless of context.

§11.4.5 — Audio + video quality analysis comprehensiveness (User mandate, 2026-05-07)

Forensic anchor — direct user mandate (verbatim, 2026-05-07):

"We MUST HAVE still analyzing of recorded materials and comprehensive validation and verification for issues we used to test! For example if there is audio at all or video, if so, is it good and proper or is it faulty? Does it have glitches, frame issues and other possible obstructions? IMPORTANT: Make sure that all existing tests and Challenges do work in anti-bluff manner — they MUST confirm that all tested codebase really works as expected!"

§11.4.2 mandates *captured* evidence; §11.4.5 mandates the **content** of that evidence be analyzed for quality, not merely for presence. A test that captures a 0-byte mp4 (Bug #24) and PASSES because "the recording file exists" is the exact PASS-bluff pattern §11.4 forbids. Content-quality analysis is what closes that gap.

Audio quality analysis — every audio test that PASSES MUST verify ALL of: (1) **Presence** — non-trivial RMS amplitude in captured WAV / `/proc/asound/.../pcm*p/sub0/hw_params`. (2) **Channel count** — `ffprobe -show_streams` matches the test's claim (2.0 / 5.1 / 7.1). (3) **Sample rate + bit depth** — match the codec / pipeline under test. (4) **Glitch census** — XRUN / FastMixer underrun-overflow-partial / AudioFlinger `writeError` counts above tolerance MUST classify explicitly (PASS within budget, WARN above, FAIL on hard limits per §11.4.1 SKIP-vs-FAIL decision tree). (5) **Coexistence-artifact census** — for tests that exercise WiFi/BT alongside audio: BT TX queue overflow, A2DP src underflow, coex notification storms, 2.4 GHz radio contention.

Video quality analysis — every video test that PASSES MUST verify ALL of: (1) **Presence** — captured screen recording has non-zero file size AND `ffprobe -count_frames` reports decoded-frame total > 0. 0-byte mp4 (Bug #24) is the canonical PASS-bluff and triggers §11.4.4 STOP. (2) **Routing target** — analyzer + action-timeline confirms video appeared on the *intended* display (primary vs secondary HDMI; Bug #13 pattern). (3) **Frame health** — drop count, frame-time variance (jitter), freeze detection (SSIM > 0.99 for ≥ 1 s), tearing. (4) **Obstruction census** — Tesseract OCR scan for hostile overlays (Application not responding, Force close, sign-in dialog, geo-restriction overlay, ad break, paywall, App is not certified). (5) **Resolution + codec** — captured frame dimensions match the test's claim; downgrade is a PASS-bluff.

Challenges (HelixQA) are bound equally — every Challenge that asserts PASS MUST run all five audio + five video layers. A Challenge that scores PASS without applicable analysis is the same class of defect as a unit test that does.

Tooling guarantee: audio = tinycap + aplay --dump-hw-params + ffprobe + /proc/asound/parsers (lib/audio_validation.sh per §11.2.5). Video = screenrecord + ffprobe -count_frames + recording-analyzer + Tesseract OCR (scripts/dual_display_record.sh

- cmd/recording-analyzer/ per §11.4.2.A and §11.4.2.C). Tests dispatched against video evidence MUST honor §11.4.4 test-interrupt-on-discovery when the analyzer reports empty input — do not silently absorb that as a generic PASS-bluff banner.

Non-compliance is a release blocker regardless of context.

MANDATORY §12 HOST-SESSION SAFETY — INCIDENT #2 ANCHOR (2026-04-28)

Second forensic incident: on 2026-04-28 18:36:35 MSK the user's user@1000.service was again SIGKILLED (status=9/KILL), this time WITHOUT a kernel OOM kill (systemd-oomd inactive, MemoryMax=infinity) — a different vector than Incident #1. Cascade killed claude, tmux, the in-flight ATMOSphere build, and 20+ npm MCP server processes. Likely cumulative cgroup pressure + external watchdog.

Mandatory safeguards effective 2026-04-28 (full text in parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §12 Incident #2):

1. scripts/build.sh MUST source lib/host_session_safety.sh and call host_check_safety BEFORE any heavy step.
2. host_check_safety has 7 distress detectors including common cgroup-events warnings (#6) and current-boot session-kill events (#7).
3. Containers MUST be clean-slate destroyed + rebuilt after any suspected §12 incident. mem_limit is per-container, not per-user-slice — operator MUST cap $\sum \text{mem_limit} \leq \text{physical RAM}$ — user-session overhead.
4. 20+ npm-spawned MCP server processes are a known memory multiplier; stop non-essential MCPs before heavy ATMOSphere work.
5. **Investigation: Docker/Podman as session-loss vector.** Per-container cgroups don't prevent cumulative user-slice pressure; common Failed to open cgroups file: /sys/fs/cgroup/memory.events warnings preceded the 18:36:35 SIGKILL by 6 min — likely correlated.

This directive applies to every owned ATMOSphere repo and every HelixQA dependency. Non-compliance is a Constitution §12 violation.

MANDATORY §12.6 MEMORY-BUDGET CEILING — 60% MAXIMUM (User mandate, 2026-04-30)

Forensic anchor — direct user mandate (verbatim):

"We had to restart this session 3rd time in a row! The system of the host stays with no RAM memory for some reason! First make sure that whatever we do through our procedures related to this project MUST NOT use more than 60% of total system memory! All processes MUST be able to function normally!"

The mandate. Project procedures MUST NOT use more than **60% of total system RAM** (HOST_SAFETY_MAX_MEM_PCT). The remaining 40% is reserved for the operator's other workloads so the host can keep serving them while project work proceeds.

Three consecutive session-loss SIGKILLS on 2026-04-30 during 1.1.5-dev — every one happened while scripts/build.sh was running m -j5 AOSP. Each Soong/Ninja job peaks at ~5–8 GiB RSS; collective RSS overran the 60% envelope and the kernel OOM-killer escalated, taking down user@1000.service. **§12.1's pre-flight check (refusing to start if host already distressed) was not enough** — the missing piece was an active CONSTRAINT on heavy work itself.

Mandatory protections (rock-solid):

1. HOST_SAFETY_MAX_MEM_PCT defaults to 60 in scripts/lib/host_session_safety.sh.
2. HOST_SAFETY_BUDGET_GB is computed at source-time from $\text{MemTotal} \times \text{MAX_PCT}/100$.
3. bounded_run clamps MemoryMax down to the budget if the caller asks for more (cgroup-level enforcement via systemd-run --user --scope -p MemoryMax=...).
4. host_safe_parallel_jobs and host_safe_build_jobs return the safe -j count given an estimated per-job RSS, capped at nproc.
5. scripts/build.sh wraps m -j in bounded_run. If the build's collective RSS exceeds the budget, only the scope is OOM-killed; user@<uid>.service stays alive.

Captured-evidence enforcement. Pre-build gate CM-MEMBUDGET-METATEST locks all 7 invariants and fires every pre-build run.

No escape hatch. §12.6 has NO operator-facing override flag. The cap exists for the operator's own protection; bypassing it is the bluff the §11.4 covenant specifically prohibits. Operators who need more headroom should reduce parallelism, close other workloads, or add RAM — NOT raise the percentage.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §12.6.

Non-compliance is a release blocker regardless of context.

MANDATORY §12.10 CONTINUATION DOCUMENT MAINTENANCE (User mandate, 2026-05-07)

Forensic anchor — direct user mandate (verbatim):

"during any work we perfrom, during Phases implementation, debugging and fixing, during ANY effort we have the Continuation document **MUST BE** maintained and it **MUST NOT BE** out of sync with current work we are doing! If for any reson we stop our work, we **MUST BE** able to continue any time, with current work, exactly where we have left of and from any CLI agent or any LLM model we chose! Nothing can be broken or faulty in maintained Continuation document!"

The mandate. A single, canonical, machine-readable handoff document — `docs/CONTINUATION.md` at the parent repo root — must always reflect the live state of the project. Any agent (human, Claude Code, Cursor, Aider, Codex, Gemini CLI, any future LLM) must be able to resume work **exactly where the previous session left off** by reading this single file.

Mandatory protections (no escape hatch):

1. `docs/CONTINUATION.md` **MUST exist** at the parent repo root.
2. **Every non-trivial state change** — work item started / completed / blocked, new bug discovered, phase transition, build state change, fix applied, gate added, mutation paired — **MUST update this document in the same commit** as the work itself. Commits that change source/tests/docs but leave `CONTINUATION.md` stale are non-compliant.
3. **Top-of-file Last updated:** ISO timestamp updated on every edit.
4. **Section §3 "Active work"** must list every IN PROGRESS / BLOCKED item with concrete commands, file paths, monitor IDs, and percentages where relevant — enough that any agent can resume without conversation context.
5. **Section §0 "How to use this document"** must contain the verbatim resumption prompt — a single block any operator can paste into any CLI agent.
6. **Document MUST be self-contained** — no hyperlinks to ephemeral external systems as the only source of truth.

Captured-evidence enforcement. Pre-build gate CM-CONTINUATION-DOC-INSYNC locks 7 invariants (file exists, has Last-updated timestamp, timestamp recent enough, has \$0 / \$3 / \$8 sections, line count ≥ 100). Pre-build gate CM-CONTINUATION-DOC-PROPAGATION verifies §12.10 text is present across parent CLAUDE.md / AGENTS.md and all 10 owned submodules' CLAUDE.md / AGENTS.md.

Commit-time enforcement. scripts/commit_all.sh refuses to commit if the staged change-set touches source / tests / gates but leaves docs/CONTINUATION.md untouched, with explicit override flag --continuation-no-update-needed (rationale captured in commit message).

Paired mutation. meta_test_false_positive_proof.sh CM-CONTINUATION-DOC-INSYNC mutation deletes the Last updated: line and asserts the gate FAILs.

No escape hatch. §12.10 has NO operator-facing override flag for the existence requirement. The discipline exists for the operator's own protection — the moment the document drifts from reality is the moment session-loss becomes catastrophic.

Canonical authority: parent ../../../../docs/guides/ATMOSPHERE_CONSTITUTION.md §12.10.

Non-compliance is a release blocker regardless of context.

§11.4.6 — No-guessing mandate (User mandate, 2026-05-08)

Forensic anchor — direct user mandate (verbatim, 2026-05-08T18:30 MSK):

"'LIKELY' is guessing, we MUST NOT have guessing, since it can be or may not be! No bluffing and uncertainty is allowed at any cost! We MUST always know exactly precisely what is happening exactly, in any context, under any conditions, everywhere!"

Tests, gates, status reports, closure narratives, commit messages, and operator-facing text MUST NOT use likely, probably, maybe, might, possibly, presumably, seems, or appears to when describing causes of failures, behaviour, or fix effectiveness. Either prove the cause with captured forensic evidence (logcat, dmesg, /sys readings, getprop, kernel ramoops, dropbox, strace, etc.) and state it as fact, OR explicitly mark UNCONFIRMED: / UNKNOWN: / PENDING_FORENSICS: with a tracked-task ID for follow-up.

Pre-build gate CM-NO-GUESSING-MANDATE greps recently-modified docs

- test scripts for the forbidden vocabulary outside explicit UNCONFIRMED: / UNKNOWN: / PENDING_FORENSICS: blocks. Paired muta-

tion introduces a likely token into a fresh status block → gate FAILs. Propagation gate CM-COVENANT-114-6-PROPAGATION enforces this anchor in every CLAUDE.md / AGENTS.md across parent + 10 owned submodules + HelixQA dependencies.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.6.

Non-compliance is a release blocker regardless of context.

§11.4.7 — Demotion-evidence rule (Phase 38.X+2 amendment, 2026-05-11)

A demotion from any FAIL classification (OPEN, POSSIBLE PRODUCT DEFECT, FAIL) to a lower-severity classification (INVESTIGATED, MITIGATED, RESOLVED, WORKING-AS-INTENDED) requires positive evidence captured under the **same conditions** that originally exposed the defect — same device, same firmware, same cycle position, same load profile.

"I cannot reproduce in isolation" is a HYPOTHESIS, not a finding. Per §11.4.6 it MUST be tagged UNCONFIRMED: until same-conditions retest produces positive evidence. The expanded forbidden-vocabulary list:

Forbidden phrase	Why it bluffs
"isolated re-run PASSes therefore X was a flake"	Strips the very environment that exposed the defect.
"runtime drift"	Label for "we don't know what changed".
"intermittent" / "transient"	Label for "we don't know how to reproduce".
"pending stress retest"	Defers the actual investigation indefinitely.
"correlates with X"	Hypothesis presented as causation.

Pre-build gate CM-DEMOTION-EVIDENCE-RULE scans Issues.md / Fixed.md / CONTINUATION.md for these phrases outside explicit UNCONFIRMED: / UNATTRIBUTED: / PENDING_CYCLE_RETEST: blocks. Propagation gate CM-COVENANT-114-7-PROPAGATION enforces this anchor in every CLAUDE.md / AGENTS.md across parent + 10 owned submodules + HelixQA dependencies.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.7.

Non-compliance is a release blocker regardless of context.

§11.4.8 — Deep-web-research-before-implementation mandate (User mandate, 2026-05-12)

Before designing a non-trivial fix, implementing a new feature, or declaring an architectural choice, perform deep web research to verify the chosen approach is informed by current state-of-the-art. Research surface: official documentation (Android/AOSP/Khronos/CEA-861/AES/IEEE/IETF/ITU), vendor technical guides (Rockchip, Sipeed, Audinate Dante, Synaptics, Realtek, Bluetooth SIG), open-source codebases (Linux kernel, ALSA, Bluez, ExoPlayer, libVLC, MPV, FFmpeg, AOSP forks), coding tutorials + technical articles (Stack Overflow, AOSP Code Lab, AES papers), issue trackers (Android bug tracker, AOSP gerit, GitHub issues).

A fix that re-invents a wheel — or reproduces a known-broken pattern — when the open-source community has already solved the problem is a §11.4 violation by omission. Every non-trivial fix's commit / Issues.md / Fixed.md entry MUST cite at least one external source URL OR the literal "NO external solution found — original work".

Pre-build gate CM-RESEARCH-CITATION-PRESENT scans new fix-direction blocks for the pattern. Propagation gate CM-COVENANT-114-8-PROPAGATION enforces this anchor in every CLAUDE.md / AGENTS.md across parent + 10 owned submodules + HelixQA dependencies.

Documentation continuity requirement: every fix landed under §11.4.8 also adds to docs/guides/ a user-facing or developer-facing guide section where appropriate.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.8.

Non-compliance is a release blocker regardless of context.

§11.4.9 — Batch-source-fixes-before-rebuild mandate (User mandate, 2026-05-12)

When closing a multi-defect batch, all source-side fixes that DO NOT require runtime on-device validation to design MUST be landed BEFORE the next firmware rebuild. Anti-pattern eliminated: Fix A → rebuild → flash → cycle → fix B → rebuild → ... serializes 7-8 hours per fix instead of batching all into ONE build cycle. Operator time is the scarce resource.

Exceptions documented in commit message as REQUIRES_REBUILD: <reason>: kernel-5.10/ changes, atmosphere-*.sh boot-script side-effects, hardware/rockchip/ HAL behavior — each gates downstream state and requires firmware to validate.

Before declaring a batch "ready for rebuild": pre-build GREEN + meta-test GREEN + existing-device validations performed where possible + Issues.md/Fixed.md/CONTINUATION.md in sync (+ HTML/PDF exported) + §11.4.8 research citations all logged.

Propagation gate CM-COVENANT-114-9-PROPAGATION enforces this anchor in every CLAUDE.md / AGENTS.md across parent + 10 owned submodules + HelixQA dependencies.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.9.

Non-compliance is a release blocker regardless of context.

§11.4.10 — Credentials-handling mandate (User mandate, 2026-05-12)

All credentials, secrets, API tokens, passwords, phone numbers, OAuth tokens, signing keys MUST NEVER live in tracked files. Templates with placeholder values are allowed (.example suffix). Tests load credentials at runtime from scripts/testing/secrets/ (or per-submodule equivalent); operator-populated files are chmod 600, directory is chmod 700. .env, .env.*, *.env patterns + scripts/testing/secrets/* (with .example + README.md exception) git-ignored project-wide.

Test scripts MUST NEVER echo credentials to stdout/stderr/logcat. Screen- recording of sign-in flows MUST redact credential-bearing frames. Per-service file separation (.netflix.env, .disney.env, etc.) limits blast radius.

Forensic-rotation policy: suspected leak → rotate at provider, update local .env, audit captured artifacts. Pre-build gate CM-CREDENTIAL-LEAK-SCAN greps tracked files for entropy-suspicious password strings + known API-token formats. Propagation gate CM-COVENANT-114-10-PROPAGATION enforces this anchor in every CLAUDE.md / AGENTS.md across parent + 10 owned submodules + HelixQA dependencies.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.10.

Non-compliance is a release blocker regardless of context.

§11.4.14 — Test playback cleanup mandate (User mandate, 2026-05-13)

Every test that issues `am start / cmd media_session play / MediaController.play` MUST issue matching `am force-stop / input keyevent KEYCODE_MEDIA_STOP` + register cleanup in EXIT trap. Verified via positive evidence (Arvus codec-state → N.E., `dumpsys media_session` shows no PLAYING for test app). `test_all_fixes.sh`

post-test sanity check FAILs the just-completed test if it left orphan playback. HelixQA Challenges bound equally. No grace period — "next test will clean it up" is §11.4 PASS-bluff.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.14. Pre-build gates CM-TEST-PLAYBACK-CLEANUP + CM-COVENANT-114-14-PROPAGATION.

Non-compliance is a release blocker regardless of context.

§11.4.15 — Item-status tracking mandate (User mandate, 2026-05-13)

Every active item in docs/Issues.md carries a ****Status:**** line with one of six values: Queued, In progress, Ready for testing, In testing, Reopened, Fixed (→ Fixed.md). Status **MUST** be updated as the item progresses through its lifecycle. Fixed requires captured-evidence per §11.4.5 + migration to Fixed.md.

The auto-generated docs/Issues_Summary.md includes the Status column. All three file types (.md, .html, .pdf) **MUST** be in sync at all times — enforced by CM-DOCS-EXPORT-SYNC (§11.4.12 + §11.4.15 amendment).

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.15. Pre-build gates CM-ITEM-STATUS-TRACKING + CM-COVENANT-114-15-PROPAGATION.

Non-compliance is a release blocker regardless of context.

§11.4.16 — Item-type tracking mandate (User mandate, 2026-05-14)

Every active item in docs/Issues.md carries a ****Type:**** line with one of three values: Bug (product defect / regression / user-visible broken behaviour), Feature (new capability not previously offered to end users), Task (internal workstream — refactor, doc, infra, gate, audit; the lowest-stakes default when ambiguous). The vocabulary is **CLOSED** — no other value is permitted.

The auto-generated docs/Issues_Summary.md includes the Type column. All three file types (.md, .html, .pdf) **MUST** be in sync at all times — enforced by CM-DOCS-EXPORT-SYNC (§11.4.12 + §11.4.15 + §11.4.16 amendment).

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.16. Pre-build gates CM-ITEM-TYPE-TRACKING + CM-COVENANT-114-16-PROPAGATION.

Non-compliance is a release blocker regardless of context.

§11.4.13 — Out-of-band sink-side captured-evidence mandate (User mandate, 2026-05-13)

Whenever an HDMI sink with a network-accessible introspection API is present (current example: Arvus H2-4D-273 at <http://192.168.4.185/>), the test suite MUST consume the sink's report as captured-evidence for every audio test asserting a codec / channel-count / passthrough mode. On-SoC HAL telemetry ALONE is insufficient — that is the exact "tests pass but the feature doesn't work" pattern §11.4 forbids. Reference: `scripts/testing/lib/arvus_probe.sh`, `scripts/testing/arvus_probe.sh`, `docs/guides/ARVUS_HDMI_INTEGRATION.md`. Pre-build gate CM-ARVUS-EVIDENCE-INTEGRATED (7 invariants) + paired mutation. No hardcoding (`env: ARVUS_HOST` etc.). Topology dispatch per §11.4.3 — sink unreachable → SKIP, never FAIL. Identity verification (MAC match) before consuming codec-state. Anti-stickiness post-stop. HelixQA Challenges bound equally.

Canonical authority: parent `docs/guides/ATMOSPHERE_CONSTITUTION.md` §11.4.13. Integration reference: `docs/guides/ARVUS_HDMI_INTEGRATION.md`.

Non-compliance is a release blocker regardless of context.

§11.4.11 — File-layout discipline (User mandate, 2026-05-12)

Files live in canonical directories per type:

- Shell scripts → `scripts/` (legacy: `scripts/legacy/`)
- Log files → `logs/` (legacy: `logs/legacy/`)
- Release artifacts → `releases/<app>/<version>/`
- Operator credentials → `scripts/testing/secrets/` (per §11.4.10, git-ignored)
- Markdown docs → `docs/` + `docs/guides/` + `docs/research/` + `docs/superpowers/plans/`
- Per-version changelogs → `docs/changelogs/`
- Hardware ID photos → `docs/hardware/<device-slug>/`

Repo root contains ONLY: AOSP-mandated top-level files (`Android.bp`, `Makefile`, `bootstrap.bash`, `BUILD`, `kokoro`, `lk_inc.mk`, `OWNERS`, `version_defaults.mk`), project metadata (`README/CLAUDE/AGENTS/CONTRIBUTING/LICENSE/NOTICE/VERSION`), dot-files (`.gitignore/.gitmodules`), and standard top-level dirs (`build/`, `device/`, `external/`, `frameworks/`, `hardware/`, `kernel-5.10/`, `packages/`, `prebuilts/`, `scripts/`, `system/`, `tools/`, `vendor/`, `docs/`, `releases/`, `logs/`).

NO bash scripts in repo root except AOSP-mandated `bootstrap.bash`. NO log files in repo root. NO duplicate filenames between root and `scripts/`. NO release artifacts in root. Moves require triple-verification (audit all references + distinguish absolute vs subdir-local + confirm no AOSP build-system requirement). Pre-build gate CM-FILE-LAYOUT-DISCIPLINE enforces.

Propagation gate CM-COVENANT-114-11-PROPAGATION enforces this anchor in every CLAUDE.md / AGENTS.md across parent + 10 owned submodules + HelixQA dependencies.

Carve-out (User mandate 2026-05-20). The 5 canonical tracker documents — docs/Issues.md, docs/Issues_Summary.md, docs/Fixed.md, docs/Fixed_Summary.md, docs/CONTINUATION.md — sit at docs/root by design. They are architectural constants of the project layout, analogous to AOSP's Makefile, Android.bp, OWNERS files at repo root. Their location is encoded as literal path strings in §11.4.12 + §11.4.15 + §11.4.16 + §11.4.19 + §11.4.44 + §11.4.53 propagation gates plus the helper-script constellation that regenerates them. Moving them would require coordinated amendment of those 6 sister anchors plus 5 pre-build gates plus ~20 helper scripts plus 42 consumer files in a single PWU. Per §11.4.66, that scope is operator-blocked until explicitly authorised. Audit-snapshot files (docs/audit/anti_bluff_audit.md, docs/audit/PRE_SONOS_TAG_READINESS.md, docs/audit/D1_WIFI_FAIL_CLASSIFICATION.md, plus any future audit snapshots) DO move under docs/audit/ per the §11.4.11 general principle.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.11.

Non-compliance is a release blocker regardless of context.

§11.4.12 — Issues_Summary.md sync mandate (User mandate, 2026-05-12)

docs/Issues_Summary.md is the canonical short-form summary of all open items. MUST be regenerated + re-exported (HTML + PDF) whenever Issues.md changes. Generator: scripts/testing/generate_issues_summary.sh. Pre-build gates CM-ISSUES-SUMMARY-SYNC + CM-COVENANT-114-12-PROPAGATION enforce mechanically.

Sort order (User mandate refinement 2026-05-12): severity DESC (C → M → L), then intra-group criticality DESC inside each group. Most critical row = #1, least critical = #N. Documented at the top of the generated file.

Auto-sync wrapper: scripts/testing/sync_issues_docs.sh — runs generator + export_progress_docs.sh in one shot. MUST be invoked after any edit to Issues.md or Issues_Summary.md. HTML+PDF exports are NEVER manually invoked; they ALWAYS travel with the markdown.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.12.

Non-compliance is a release blocker regardless of context.

§11.4.33 — Type-aware closure-status vocabulary (User mandate, 2026-05-15)

§11.4.15 defined the lifecycle Status closed-set including terminal Fixed ($\rightarrow$ Fixed.md). §11.4.16 defined the Type closed-set {Bug | Feature | Task}. §11.4.33 binds the two — closure terminal value MUST agree with the item Type: Bug $\rightarrow$ Fixed ($\rightarrow$ Fixed.md), Feature $\rightarrow$ Implemented ($\rightarrow$ Fixed.md), Task $\rightarrow$ Completed ($\rightarrow$ Fixed.md). The ($\rightarrow$ Fixed.md) suffix is preserved so existing migration tooling (atomic Issues.md $\rightarrow$ Fixed.md move per §11.4.19) keeps working. Generators treat the three terminal values as semantically equivalent (all closed, positive evidence captured) but preserve the literal in emitted docs. Closing a Feature with Fixed ($\rightarrow$ Fixed.md) or a Task with Implemented ($\rightarrow$ Fixed.md) is a §11.4.33 violation. Pre-build gate CM-CLOSURE-VOCAB-TYPE-AWARE.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.33.

Non-compliance is a release blocker regardless of context.

§11.4.34 — Reopened-source attribution mandate (User mandate, 2026-05-15)

Every Issues.md heading whose ****Status:**** is Reopened MUST carry a ****Reopened-Details:**** line within 8 non-blank lines of the heading, capturing four sub-facts: **By:** AI or User; **On:** ISO date; **Reason:** one of { test-failed | manual-testing-detected | captured-evidence-contradicts | end-user-report | cycle-re-discovered | design-reconsidered } or explicit free text; **Evidence:** path or short description of the captured artefact. Reopens without evidence are §11.4.6 / §11.4.7 violations: the reopen IS a demotion-from-Fixed change. Issues_Summary.md Status column MUST distinguish Reopened sub-states by source (e.g. Reopened (AI: test-failed) vs Reopened (User: manual-testing)). Pre-build gate CM-ITEM-REOPENED-DETAILS mirrors CM-ITEM-OPERATOR-BLOCKED-DETAILS walk pattern.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.34.

Non-compliance is a release blocker regardless of context.

§11.4.35 — Canonical-root inheritance clarity (User mandate, 2026-05-15)

The constitution submodule's three files (constitution/Constitution.md, constitution/CLAUDE.md, constitution/AGENTS.md) ARE the canonical root — also called the parent files. Universal rules per §11.4.17 live there.

This submodule's CLAUDE.md and AGENTS.md are consumer extensions of the parent ATMOSphere-Android-15 project, which is itself a consumer extension of the constitution submodule. Both layers open with an inheritance pointer (the `## INHERITED FROM` heading near the top of each, or `@constitution/CLAUDE.md` for agents that resolve `@imports`). This submodule's files contain only rules specific to this submodule's role inside ATMOSphere (player wiring, on-device test contract, build-step integration, APK module name, applicationId conventions). Project-wide ATMOSphere rules live in the parent ATMOSphere-Android-15 CLAUDE.md / AGENTS.md. Universal rules live in the constitution submodule.

When in doubt: universal rule → constitution submodule; ATMOSphere-wide rule → parent project repo root; this-submodule-only rule → this file. Default to the narrowest layer when uncertain. "Parent CLAUDE.md" / "root Constitution" → constitution submodule file at `constitution/<filename>`, NEVER this submodule's own CLAUDE.md / AGENTS.md and NEVER the parent ATMOSphere-Android-15 project root files. Moving a rule between layers MUST be a visible commit — `git mv` + an explicit "Lifted from to per §11.4.35" line in the message. AI agents MUST NOT silently re-author a §11.4.X anchor in the wrong layer and call it propagation.

Recommended pre-build gate CM-CANONICAL-ROOT-CLARITY verifies the inheritance pointer is present and the constitution submodule files are reachable. Composes with §11.4.17.

Canonical authority: constitution submodule Constitution.md §11.4.35.

Non-compliance is a release blocker regardless of context.

§11.4.40 — Full-suite retest before release tag mandate (User mandate, 2026-05-17)

A release tag MUST NOT be created until a COMPLETE retest with ALL existing tests has been executed on a clean baseline AFTER every workable item in the batch is done, fixed, polished, and individually verified. Spot-check retests that run only the tests touched by the batch are FORBIDDEN — they miss interaction defects between the batch's fixes and previously-stable code.

The complete retest comprises: (1) pre-build full sweep, (2) post-build full sweep, (3) on-device 4-phase cycle on EVERY owned device, (4) meta-test full mutation sweep, (5) Challenge bank full sweep, (6) Issues.md/Fixed.md state audit, (7) CONTINUATION.md sync check.

Time is essential — complete retest is typically 12-48 hour elapsed effort. NOT optional, NOT abbreviated. Skipping is the exact "tests passed but feature broken" failure mode §11.4 specifically prohibits.

Composes with §11.4.4 (per-fix retest) — §11.4.37 is the additional final integrity check at RELEASE granularity. Composes with §11.4.7 — full-suite retest is the authoritative baseline for closures in the batch. No escape hatch — no --skip-full-retest or --quick-release flag exists.

Pre-build gate CM-FULL-SUITE-RETEST-MANDATE + paired mutation. Propagation gate CM-COVENANT-114-40-PROPAGATION enforces this anchor in every CLAUDE.md/AGENTS.md across parent + 10 owned submodules + HelixQA dependencies.

Canonical authority: constitution submodule Constitution.md §11.4.37.

Non-compliance is a release blocker regardless of context.

§11.4.41 — Pre-Force-Push Merge-First Mandate (User mandate, 2026-05-17)

Any force-push (git push --force, git push --force-with-lease, git push +<ref>, or equivalent history-rewriting operation on any remote) authorised under §9.2 / CONST-043 MUST be preceded by a mechanical 4-step merge-first pipeline that brings every remote-side commit into the local tree, resolves every conflict carefully, and verifies nothing is lost or corrupted on EITHER side BEFORE the overwriting push is executed.

The 4-step pipeline (mandatory, in order): (1) git fetch --all --prune --tags against every configured remote — capture output. (2) Integrate every divergent commit locally via git rebase (local is strict superset), git merge (independent additions both deserve preservation), or operator-confirmed cherry-pick (remote subset already present locally). (3) Audit: no conflict markers (grep -rn '^<<<<<< \|^===== \$\|^>>>>>>' returns empty), no silent file drops (git diff --stat HEAD@{1} HEAD), every previously-passing test still passes per §11.4.4 / §11.4.40 baseline, every captured-evidence artifact still validates. (4) git push --force-with-lease <remote> <ref> (NEVER --force without --with-lease unless §9.2 sub-clause 6 explicitly authorises it for a remote where lease semantics are unavailable). One force-push event per CONST-043 authorisation — no batch authorisation.

Two-gate composition with CONST-043 — §11.4.41 does NOT relax CONST-043's operator-approval requirement. Gate A (CONST-043): operator types explicit per-operation force-push authorisation. Gate B (§11.4.41): agent executes the 4-step merge-

first pipeline, captures evidence of clean integration, presents evidence to operator BEFORE the force-push. Both gates required.

Verification artefact — every §11.4.41-governed force-push emits a docs/changelogs/<tag>.md "Force-push merge-first audit" section containing 7 elements: (i) git fetch output, (ii) per-remote HEAD..<remote>/<branch> log before integration, (iii) integration strategy chosen per remote with rationale, (iv) post-integration conflict-marker scan output (must be empty), (v) post-integration test suite delta (must show only expected changes), (vi) --force-with-lease push output with lease SHA evidence, (vii) CONST-043 authorisation quote from the conversation.

Composes with §9.2 (data-safety hardlinked backup), §11.4.4 (test-interrupt-on-discovery — broken integration triggers rollback), §11.4.6 (no-guessing — every step's outcome captured, not assumed), §11.4.26 (constitution-submodule update pipeline — per-submodule specialisation), §11.4.32 (post-pull validation — audit step's mechanical companion), §11.4.37 (fetch-before-edit — step 1 enforces it for force-push specifically), §11.4.40 (full-suite retest — step 3's test-evidence requirement).

No escape hatch — the operator-pressure escape ("just force-push, we'll fix it later") is the exact failure mode this anchor closes. Pre-build gate CM-COVENANT-114-41-PROPAGATION enforces this anchor in every CLAUDE.md/AGENTS.md across parent + 10 owned submodules + nested submodules + HelixQA dependencies. Paired mutation strips the anchor literal → gate FAILs. Gate CM-FORCE-PUSH-MERGE-FIRST walks docs/changelogs/<tag>.md "Force-push" entries for the 7 audit elements; paired mutation strips any element and asserts gate FAILs.

Canonical authority: constitution submodule Constitution.md §11.4.41.

Non-compliance is a release blocker regardless of context.

§11.4.52 — Autonomous-Validation Mandate (User mandate, 2026-05-18)

Forensic anchor — verbatim user mandate (2026-05-18):

"Make sure we have full automation tests which will do all this work in full automation! IMPORTANT: Make sure that all existing tests and Challenges do work in anti-bluff manner — they MUST confirm that all tested codebase really works as expected! execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product! This MUST BE part of Constitution of our project, its CLAUDE.MD and AGENTS.MD if it is not there already, and to be applied to all Submodules's Constitution, CLAUDE.MD and AGENTS.MD as well."

Every user-facing feature **MUST** have at least one autonomous validation path: end-to-end via `adb shell` + scripted automation, captured runtime evidence per §11.4.5, PASS/FAIL verdict **WITHOUT** human presence to drive UI, observe screen, or make decisions. Operator-attended tests are **SUPPLEMENTARY**, never **PRIMARY**. A feature whose **ONLY** validation path is operator-attended is a §11.4.52 violation — the path does not scale to CI, does not run on every commit, does not survive operator unavailability, and produces the exact "tests pass but feature doesn't work for users" failure mode §11.4 forbids.

Acceptable autonomous paths: (a) programmatic instrumentation APK (SDK-API exercises like `MediaCodec.createDecoderByName` + structured JSON result file); (b) headless intent dispatch + state poll (`am start -es / am broadcast + dumsys /proc/<pid>/maps / media.metrics` polling); (c) ADB-driven `uiautomator` (**ONLY** if hierarchy has ≥ 1 clickable node — empty hierarchy demands fallback to APK/intent); (d) network-side sink probe per §11.4.13; (e) HelixQA autonomous QA session per §11.4.27.

Coverage ledger (§11.4.25) classifies each feature as `AUTONOMOUS_VERIFIED` / `AUTONOMOUS_DESIGNED` / `OPERATOR_ATTENDED_ONLY` / `NOT_APPLICABLE`. `OPERATOR_ATTENDED_ONLY` blocks release until migrated; cite tracked work item per §11.4.15 + §11.4.16. Autonomous paths themselves **MUST** be anti-bluff: positive captured evidence + paired meta-test mutation per §1.1.

Composes with §11.4.25 (full-automation coverage), §11.4.27 (no-fakes + 100% type coverage), §11.4.39 (per-feature on-device end-user validation), §11.4.43 (TDD RED-first), §11.4.48 (UI-driven — fallback to APK/intent when `uiautomator` hierarchy empty), §11.4.49 (dual-approach), §11.4.50 (deterministic consistency), §11.4.51 (live-ADB-first).

Pre-build gates: `CM-COVENANT-114-52-PROPAGATION` + `CM-AF-AUTONOMOUS-PATH-PER-FEATURE`. Paired mutations. No escape hatch — no `--allow-operator-attended-only`, `--skip-autonomous-path`, `--manual-validation-suffices` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.52.

Non-compliance is a release blocker regardless of context.

§11.4.53 — Fixed_Summary parity mandate (User mandate, 2026-05-18)

Forensic anchor — verbatim user mandate

(2026-05-18T17:55Z): "Note: Just like for Issues we have `Issues_Summary`, for Fixed we **MUST HAVE** `Fixed_Summary` - like all other docs: **ALWAYS** in sync and up to date and **ALWAYS** expor-

ted into the PDF and HTML! Add this mandatory rule / constraint into the root (constitution Submodule) Constitution, AGENTS.MD and CLAUDE.MD."

docs/Fixed_Summary.md is the symmetric short-form summary of docs/Fixed.md. MUST be regenerated whenever Fixed.md changes. HTML + PDF exports MUST travel with the markdown (identical mtimes within sync_issues_docs.sh granularity). Stale exports are §11.4.53 violations regardless of whether the underlying .md is correct. Same discipline as §11.4.12 Issues_Summary applied to Fixed.md.

Generator: scripts/testing/generate_fixed_summary.sh (canonical, executable, emits markdown table with Status + Type columns per §11.4.19 column-alignment). Auto-sync wrapper: scripts/testing/sync_issues_docs.sh regenerates BOTH summaries in one shot, exports HTML + PDF, colorizes per §11.4.23, re-renders PDFs. MUST be invoked after any edit to Fixed.md. No --issues-only flag exists, and §11.4.53 prohibits adding one.

Sort order: closure date DESC (most-recent-Fixed first), \$-letter / Fix-# secondary. Documented at the top of the generated file.

Composes with §11.4.12 (Issues_Summary sibling — canonical pair), §11.4.19 (atomic Issues→Fixed migration trigger + column-alignment), §11.4.23 (colorizer post-processes both summaries), §11.4.33 (type-aware closure vocabulary — Fixed_Summary respects Fixed (→ Fixed.md) / Implemented (→ Fixed.md) / Completed (→ Fixed.md) terminal values), §11.4.44 (revision header applies to Fixed_Summary.md), §12.10 (CONTINUATION.md resumption guarantee).

Pre-build gates: CM-FIXED-SUMMARY-SYNC (6 invariants — Fixed_Summary exists + HTML/PDF mtime ≥ md mtime + Fixed_Summary mtime ≥ Fixed mtime + generator + sync wrapper invokes generator) + CM-COVENANT-114-53-PROPAGATION (anchor literal across canonical files). Paired mutations strip the anchor literal AND move the generator aside AND backdate Fixed_Summary mtime. No escape hatch — no --skip-fixed-summary-sync, --issues-only, --summary-not-applicable flag.

Canonical authority: constitution submodule Constitution.md §11.4.53.

Non-compliance is a release blocker regardless of context.

§11.4.58 — Parallel-development methodology (User mandate, 2026-05-19)

Project work proceeds through the **Parallel Work Unit (PWU) pipeline** rather than sequential Phase-chain. Each PWU has: ATM-NNN identifier (§11.4.54), Issues.md entry (§11.4.15+§11.4.16), file-scope manifest, §11.4.43 RED test,

source patch, pre-build gate, post-flash test, paired §1.1 meta-test mutation, HelixQA Challenge bank entry, captured-evidence directory (§11.4.5+§11.4.52).

5-stage pipeline: Stage 1 DEVELOP (parallel PWU agents in worktrees) → Stage 2 MERGE (serial conductor + §11.4.41 4-step merge-first) → Stage 3 REBUILD+FLASH (parallel where hardware allows) → Stage 4 VALIDATE (parallel D3+D4+meta-test+coverage) → Stage 5 SWEEP (parallel HelixQA + Fixed.md migration + README refresh). Stage 1 of round N+1 overlaps with Stages 4-5 of round N.

Synchronization: 4-layer lock hierarchy (parent flock / per-submodule git / contention-path advisory locks for 10 forbidden cross-PWU paths / per-PWU worktree). Disjoint-scope PWUs fully parallel.

Anti-bluff merge-time enforcement (mandatory, all four): C1 §11.4.43 RED-test captured. C2 §1.1 paired meta-test mutation FAILs the gate. C3 §11.4.50 3-iter (or 10-iter) deterministic-consistency. C4 §11.4.5 captured-evidence per feature type. Metadata-only / configuration-only / absence-of-error / grep-without-runtime PASS REJECTED. HelixQA Challenge bank coverage MANDATORY for every user-visible PWU.

Phase 39.EX infrastructure gates (5 gates land the parallel infrastructure itself): CM-PWU-PARALLEL-VALIDATION-ORCHESTRATOR, CM-PWU-HELIXQA-PER-DOMAIN-RUNNER, CM-PWU-WORKER-POOL-LOCKING, CM-PWU-FILE-SCOPE-PARTITION, CM-PWU-AUTO-MERGE-GATE-6CONDITIONS. Each ships a paired meta-test mutation per §1.1.

Pre-build gates CM-PWU-LOCK-HIERARCHY + CM-PWU-ANTI-BLUFF-COVERAGE

- CM-PWU-MERGE-QUEUE-DISCIPLINE + CM-PWU-PARALLEL-AGENT-LIMIT + CM-COVENANT-114-58-PROPAGATION. Paired mutations cover each gate. No escape hatch.

Canonical authority: constitution submodule Constitution.md
§11.4.58. Project-specific implementation reference: docs/guides/PARALLEL_DEVELOPMENT_METHODODOLOGY.md.

Non-compliance is a release blocker regardless of context.

§11.4.65 — Universal Markdown export mandate (User mandate, 2026-05-19)

Every Markdown document inside the project that is NOT part of an application or service's source-code tree MUST have synchronized .html and .pdf siblings. Includes: project-root *.md, docs/**/*.md, scripts/**/*.*md (doc-format companion docs), owned-submodule top-level README.md / CLAUDE.md / AGENTS.md / CHANGELOG.md and their docs/**/*.*md, constitution/**/*.*md, owned HelixQA submodules' equivalents. Excludes: external/**,

prebuilts/**, packages/modules/**, kernel-5.10/**, out/**, build/**, application/service source-code trees, and third-party submodules NOT in the owned set. Every edit triggers regeneration via scripts/testing/sync_all_markdown_exports.sh (pandoc HTML + weasyprint PDF, timeout 60 per file, capped at 500 candidates). HTML + PDF mtime MUST be $\geq$ source .md mtime at all times.

Pre-build gates CM-UNIVERSAL-MARKDOWN-EXPORT-SYNC + CM-COVENANT-114-65-PROPAGATION. Paired meta-test mutations. Composes with §11.4.12 / §11.4.18 / §11.4.23 / §11.4.44 / §11.4.45 / §11.4.53 / §11.4.59 / §11.4.60 / §11.4.63 / §11.4.64. No escape hatch — no --skip-md-exports, --no-pdf-only, --md-export-not-applicable flag.

Canonical authority: constitution submodule Constitution.md §11.4.65.

Non-compliance is a release blocker regardless of context.

§11.4.66 — Blocker-resolution interactive-clarification mandate (User mandate, 2026-05-19)

When any task is blocked (operator decision, hardware access, external authorization, ambiguous scope), the agent MUST: (1) research what's doable from the agent side without operator input; (2) calculate minimum-viable operator input; (3) construct 2–4 mutually-exclusive options with one marked "Recommended" and each stating what the agent does after that answer; (4) present via the platform's interactive question mechanism (AskUserQuestion on Claude Code) — NEVER free-text "what would you like?" for closed-set decisions; (5) after the answer, resume work without follow-up round-trips. Composes with §11.4.6 / §11.4.7 / §11.4.40 / §11.4.41 / §11.4.42 / §11.4.52. No silent waiting; no bulk-text questions when interactive options would do.

Pre-build gate CM-COVENANT-114-66-PROPAGATION enforces the anchor literal across the 42-file consumer fleet. Paired meta-test mutation strips the literal → gate FAILs. No escape hatch — no --skip-ask, --silent-wait, --free-form-only flag.

Canonical authority: constitution submodule Constitution.md §11.4.66.

Non-compliance is a release blocker regardless of context.

§11.4.67 — Shell-script target-shell-parseability mandate (User mandate, 2026-05-19)

Forensic anchor — direct user mandate (verbatim, 2026-05-19): "any issue we spot must be fixed, bash scripts as well if they are broken!"

- "Make sure that this is mandatory rule!"

Every shell script that may be invoked under a target shell other than the one in its shebang MUST parse cleanly under that target shell. Forensic incident: device/rockchip/rk3588/tests/test_all_fixes.sh:114 used bash-only `exec > >(tee -a "$f") 2>&1` on a `sh` script. `sh` callsite — Android `mksh` parses the whole script BEFORE executing, so the runtime `[ -n "${BASH_VERSION:-}" ]` guard could not save it. Fixed by wrapping in `eval 'exec > >(tee ...) 2>&1'` so the parser sees only a string.

Closed-set scope: every tracked `.sh` under `device/rockchip/rk3588/tests/`, `scripts/`, `scripts/testing/` (and equivalent paths in owned submodules). OUT of scope: `external/`, `prebuilts/`, `packages/modules/`, `kernel-5.10/`, `out/`, `build/`, `scripts/legacy/`. Mandatory invariants: (1) every in-scope script parses under `sh -n`; (2) bash-only constructs (`>(...)`, `<(...)`, `[[ ]]`, `<<<`, arrays, `${var^^}`, etc.) MUST be wrapped in `eval` OR guarded by bash-only loading; (3) shebangs honest — `#!/bin/bash` only if bash actually expected; (4) fix at source per §11.4.1, never at callsites. Composes with §11.4.1 / §11.4.4 / §11.4.6 / §11.4.50 / §11.4.51.

Pre-build gate `CM-SCRIPT-TARGET-SHELL-PARSEABLE` runs `sh -n` on every in-scope script. Propagation gate `CM-COVENANT-114-67-PROPAGATION` enforces the anchor literal across the 44-file consumer fleet. Paired mutations: inject bash-only outside `eval` → parse gate FAILs; strip 11.4.67 literal → propagation gate FAILs. No escape hatch — no `--skip-parseability-check`, `--bash-only-script`, `--runtime-guard-suffices` flag.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.67.

Non-compliance is a release blocker regardless of context.

§11.4.69 — Universal sink-side positive-evidence taxonomy + mechanical enforcement (User mandate, 2026-05-20)

Forensic anchor — direct user mandate (verbatim, 2026-05-20):

"THIS MUST HAPPEN NEVER AGAIN!!! We MUST HAVE this all working! Not just for audio but for every single piece of the System!!! Proper full automation when executed with success MUST MEAN that manual testing will be as much positive at least regarding the success results! ... Solution MUST BE universal, generic that solves working flows for all System components and for all future and all existing pro-

jects! ... Everything we do MUST BE validated and verified with rock-solid proofs and anti-bluff policy enforcement and fulfillment!"

Universal generalisation of §11.4.68 (audio-specific) across every user-visible feature class. Closes the PASS-bluff pattern where tests reported green while end users hit broken features (2026-05-19→20 D3 audio "82/84 PASS" + empty Arvus Codec-In-Use).

The mandate. Every user-visible feature MUST map to one entry in the closed-set §11.4.69 sink-side evidence taxonomy (audio_output, audio_input, video_display, network_throughput, network_connectivity, bluetooth_a2dp, bluetooth_pair, touch_input, sensor, gpu_render, storage_read, storage_write, mediaco-dec_decode, mediacodec_encode, miracast, cast, boot_service, package_install, permission_grant, wifi_link, wifi_throughput, ethernet_link, display_topology, drm_playback, subtitle_render — open to additions). Every PASS for a feature in the taxonomy MUST cite a captured-evidence artefact path matching the required evidence shape.

Helper contracts (additive during grace; mandatory after 2026-06-19):

- `ab_pass_with_evidence` <description> <evidence_path> — the new canonical PASS helper. Verifies path exists AND non-empty; emits PASS: <description> [evidence: <path>].
- `ab_skip_with_reason` <description> <closed-set-reason> — reasons: `geo_restricted`, `operator_attended`, `hardware_not_present`, `topology_unsupported`, `network_unreachable_external`, `feature_disabled_by_config`. Forbids `network_unreachable_external` for any taxonomy feature with a sink-side probe.
- Bare `ab_pass` deprecated — WARN pre-grace, FAIL post-grace (2026-06-19).

Mechanical enforcement. Three pre-build gates + three paired §1.1 meta-test mutations:

- CM-SINK-EVIDENCE-PER-FEATURE — walks tests for # §11.4.69 FEATURE: <class> annotation + verifies taxonomy probe + `ab_pass_with_evidence` use.
- CM-NO-FAIL-OPEN-SKIP — audits sink-side probe helpers; FAILs if any code path converts empty/unreachable response to PASS-counting SKIP for a feature class with a sink-side probe.
- CM-AB-PASS-WITH-EVIDENCE-EVERYWHERE — pre-grace WARN, post-grace FAIL on bare `ab_pass` calls.

Composes with §11.4.1 (FAIL-bluffs forbidden), §11.4.2 (recorded-evidence), §11.4.5 (audio + video 5-layer quality), §11.4.6 (no-guessing), §11.4.13 (sink-side captured-evidence), §11.4.27 (no-

fakes-beyond-unit), §11.4.50 (deterministic consistency), §11.4.52 (autonomous-validation), §11.4.68 (audio-specific sink-side — §11.4.69 is the universal generalisation).

No escape hatch — no --skip-evidence, --config-only-pass, --allow-fail-open-skip, --legacy-ab-pass-permitted flag. The discipline exists because the 2026-05-20 forensic incident demonstrated the failure: tests reported audio-routing PASS while the user heard nothing and the Arvus Codec-In-Use field was empty.

Propagation gate CM-COVENANT-114-69-PROPAGATION enforces this anchor literal across the ~44-file consumer fleet. Paired mutation strips the literal → gate FAILs.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.69.

Non-compliance is a release blocker regardless of context.